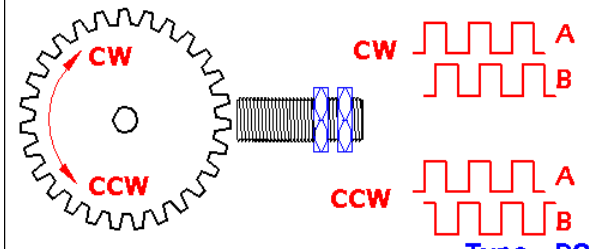


S56FW-37ADQ-5KCP4 - Quadrature (2 Channel) Gear Tooth Sensor

www.sensorso.com

Quadrature Speed Sensor, True 0 speed, 110C, 25G hys, npn 5k pull up resistor, Stainless 9/16-32 x 2.34" with face seal housing, 4 pin Fem. Weather Pac with 1 ft jacket

Two (2) Outputs/Tooth,
90° Out of Phase



- o TRUE ZERO SPEED
- o ALL STEEL GEARS 4-24 PITCH
- o TEMPERATURE STABLE

ENVIRONMENTAL SPECIFICATIONS - S56FW	
Corrosion Resistance	500 hours salt spray ASTM B-117
Installation Torque	25 Foot-Pounds Maximum
Enclosure	Nema 1,3,4,6,13 & IEC IP67
Vibration	10 G's 2 to 2000 Hz Sinusoidal
Mechanical Shock	100 G's, 11 ms Half-Sine
Pressure at face	>> 500 PSI (limits of test)

S56FW, Housing, Stainless Steel, 9/16-32, w/Washer & Nut

9/16-32 UNR-2A FITTING PER SAE J514h

SOLID FACE THICKER THAN STANDARD HOUSING FACE REDUCES SENSING GAP BY .013"

.482^{+0.020}_{-0.030}

O-RING .070 X .426ID VITON90

2.34

1.59

.84

1.25

HOUSING MATERIAL: 303 STAINLESS STEEL 2 X ALIGN KNOTCH CONNECTION SIDE

5/8" (.625) HEX

NUT 3/4" HEX X 1/2" THICK STAINLESS STEEL

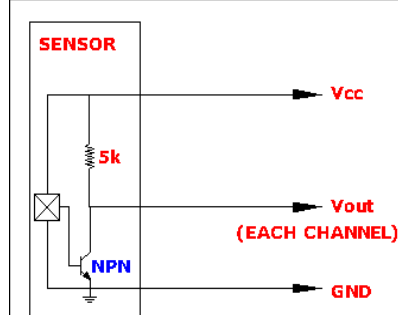
WASHER (.0791" X .035" THK) PER SAE J1926/3 5.4

Ø.600

Rev B

DIM=INCH, ID=.452

5K, 5k Pull-up Resistor



CP4, 4 Pin Female Weather Pac w/ 1FT Jacket

CONNECTOR: PACKARD WEATHER PAC #1201-5797

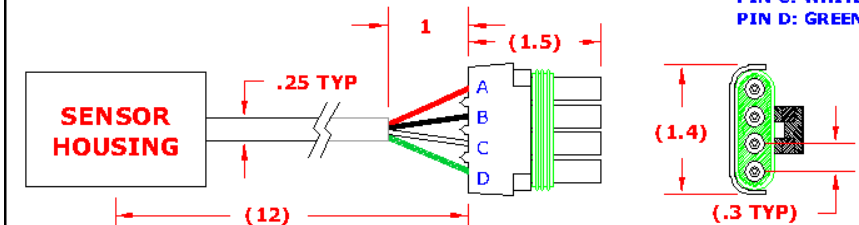
JACKETED CABLE: 4 WIRE 18 AWG PVC, 105°C

PIN A: RED

PIN B: BLACK

PIN C: WHITE

PIN D: GREEN



CHOOSE FROM CP4 MATING CONNECTORS AND CABLE SETS

Rev C

DIM=INCH

Connections Chart

Pin A Red	Vcc
Pin B Black	Ground
Pin C White	Channel A
Pin D Green	Channel B

CP4-37ADQ

Date Code 'YYM'

YY = YEAR, M = MONTH

A JAN	D APR	H JUL	L OCT
B FEB	E MAY	J AUG	M NOV
C MAR	G JUN	K SEP	N DEC

The 37ADQ Quadrature Speed Sensor provides 2 digital outputs as gear teeth rotate by the sensor's face. These sensors have 2 internal sensing elements that are physically placed 1/4, 3/4 or 1 1/4 tooth apart. The quadrature phase angle (phase shift from channel A to channel B) ideally should be 90 degrees. This will only happen if the correct gear pitch is used, and the sensor is aligned correctly with the target's rotation direction. Therefore, when ordering this sensor, it cannot be built until the factory receives the gear pitch for the intended target.

Gear pitch = (the # of teeth on the gear) / (the gear's contact diameter in inches)

Specifications may not be valid when the application is not detecting an end-sensed, 14.5 pressure angle, steel spur gear. The presence of ferrous metals or strong magnetic fields in close proximity to the sensor's internal magnet may also invalidate the specifications. Engineers are available to assist in target design and applications with non-standard targets. Custom target specifications can only be guaranteed when the customer supplies a target along with any additional components that may affect sensor output, and the customer has validated function in the finished application.

The 37ADQ sensor is a true zero speed Hall Effect Gear Tooth sensor which will never miss a pulse, and is the best choice for detecting 4-24 pitch gears with evenly spaced teeth. The 37ADQO sensor is another option for quadrature speed sensing. This sensor is the best choice for detecting gears with missing or unevenly spaced teeth, as well as other targets such as bolt heads or holes in a rotating disc. Check our website or call us to discuss all of our dual output gear tooth and quadrature speed sensor options.

Sensor Solutions * V:(970) 879-9900 F:(970) 879-9700 * www.sensorso.com * Rev ACC

S56FW-37ADQ-5KCP4 - Quadrature (2 Channel) Gear Tooth Sensor

www.sensorso.com

Quadrature Speed Sensor, True 0 speed, 110C, 25G hys, npn 5k pull up resistor, Stainless 9/16-32 x 2.34" with face seal housing, 4 pin Fem. Weather Pac with 1 ft jacket

37ADQ-5K, Electrical & Functional Specifications

ABSOLUTE MAX LIMITS	MIN	MAX	UNITS
Supply Voltage, Vcc	-15	+28	Volts DC
Voltage applied to output	-0.3	+28	Volts
Current into output	--	30	mA
Current out of output	--	Vcc/5k	mA
Load Dump, 40mS Rs = 20	--	60	Volts

ELECTRICAL SPECS	CONDITIONS	MIN	MAX	UNITS
Temperature Range *	Operating	-40	+110	Deg C
Supply Voltage, Vcc	Over temperature	+4.5	+24	Volts DC
Supply Current, Output Off	Into Vcc	+4	+15	mA
Frequency Range	True zero speed	0	10	kHz
Vol, Low Level Vout	I sink = 20 mA	0	0.4	Volts
Internal pull up resistor	From Vcc to Vout	4.95	5.05	k Ohms
Output Rise Time 10-90%	C < 100pF	--	3.0	uS
Output Fall Time 90-10%	C < 100pF	--	1.0	uS
ESD **	Nondestructive	--	2000	Volts
EMI **	20k to 1 G Hz	--	20	V/M

* T max = 150°C is available, contact factory.

** Similar Product Qualified

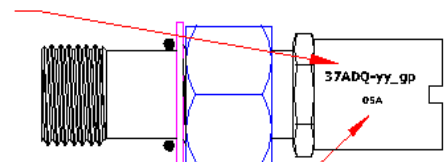
TARGET PERFORMANCE GEAR PITCH ~(#Teeth/Diam. in inches)	AIR GAP RANGE***	TYPICAL MAX GAP	TYP. OUTPUT DUTY CYCLE	ALIGNMENT SKEW ANGLE
4 (.785" tooth to tooth)	.000 to .140"	.160"	40 to 60%	±35 deg
6 (.524" tooth to tooth)	.000 to .120"	.140"	40 to 60%	±30 deg
8 (.393" tooth to tooth)	.000 to .100"	.125"	40 to 60%	±30 deg
12 (.262" tooth to tooth)	.000 to .075"	.100"	35 to 65%	±25 deg
16 (.196" tooth to tooth)	.000 to .055"	.075"	35 to 65%	±20 deg
20 (.157" tooth to tooth)	.000 to .035"	.050"	30 to 70%	±15 deg
24 (.131" tooth to tooth)	.000 to .020"	.030"	30 to 70%	±10 deg

Rev A

*** Non contacting

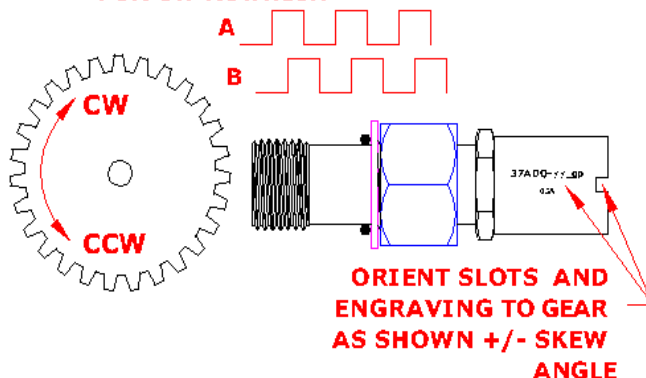
*** Phase Angle 5° to 175° over entire air gap range

**CHARACTERISTIC-OPTION_GEAR PITCH
MARKED ON THIS SURFACE, yy=OPTION,
gp=GEAR PITCH (e.g. _08 = 8 PITCH GEAR)**

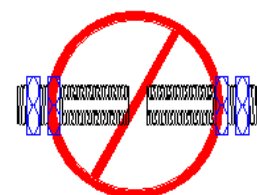


**DATE CODE,
THIS SURFACE**

**ALIGNED AS SHOWN, CH A LEADS CH B
FOR CW ROTATION**



**DO NOT CONTACT
FACE TO FACE**



**CONTACT WITH OTHER
MAGNETS MAY REDUCE THE
MAXIMUM OPERATING GAP**

S56FW-37ADQ